

Functions

Worksheet 1 (Non-Calculator)

1. S18/qp12/q11

$$f(x) = \frac{1}{3x + 2}$$

- a. Find $f(-2)$. [1]
- b. Find $f^{-1}(x)$. [2]

2. W18/qp11/q6

$$f(x) = \frac{3}{x + 4}$$

- a. Find $f(-6)$. [1]
- b. Find $f^{-1}(x)$. [2]

3. W18/qp12/q6

$$f(x) = \frac{2x + 5}{3x}$$

- a. Find $f(-2)$. [1]
- b. Find $f^{-1}(x)$. [3]

4. W19/qp11/q19

$$f(x) = \frac{5 - x}{x}$$

- a. Find $f\left(\frac{1}{2}\right)$. [1]
- b. Find $f^{-1}(x)$. [2]

5. S20/qp12/q22

$$f(x) = 4(3 - x) \qquad g(x) = \frac{5(3x - 2)}{x}$$

- a. Find $f^{-1}(x)$. [2]
- b. Find $g^{-1}(x)$. [3]

Answer Key

1. S18/qp12/q11

11(a)	$-\frac{1}{4}$ oe	1	
11(b)	$\frac{1-2x}{3x}$ oe final answer	2	M1 for correct first step: $y(3x+2)=1$ or $x=\frac{1}{3y+2}$ or $3x+2=\frac{1}{y}$ or better

2. W18/qp11/q6

6(a)	-1.5 oe	1	
6(b)	$\frac{3-4x}{x}$ oe final answer	2	B1 for $y(x+4)=3$; or $x=\frac{3}{y+4}$; or better

3. W18/qp12/q6

6(a)	$-\frac{1}{6}$ oe	1	
6(b)	$\frac{5}{3x-2}$ oe final answer	3	B2 for $3yx-2y=5$; or for $x(3y-2)=5$ or B1 for $3xy=2x+5$; or for $x=\frac{2y+5}{3y}$; or for $3xy-2x=5$; or for $3xy=2y+5$

4. W19/qp11/q19

19(a)	9	1	
19(b)	$\frac{5}{x+1}$	2	B1 for $xy=5-x$ or for $x=\frac{5-y}{y}$ or for $\frac{5}{x}=y+1$ or better

5. S20/qp12/q22

22(a)	$\frac{12-x}{4}$ or $3-\frac{x}{4}$ final answer	2	M1 for $x=4(3-y)$ or $\frac{y}{4}=3-x$ or $4x=12-y$
22(b)	$x=\frac{10}{9}$ oe	3	M1 for $15x-10=6x$ or $5(3x-2)=6x$ M1 for $9x=10$ or $ax=b$ FT their first step